

Program: Diploma in Artificial Intelligence & Machine Learning	
Course Code: 5349A	Course Title: Natural Language Processing Lab
Semester: 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Learn the techniques in natural language processing.
- Familiar with the natural language generation.
- Understand machine translation and speech recognition.

Course Prerequisites:

Topic	Course code	Course name	Semester
Sound knowledge of Python programming	1008	Introduction to IT systems Lab	I
Sound knowledge of Python programming	3347	Python Programming Lab	III
Knowledge of data sets	4347	Machine Learning Lab	IV

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Develop Programs for Word Analysis	12	Applying
CO2	Develop programs for word generation	12	Applying
CO3	Implement programs for Morphological Analysis	12	Applying

CO4	Open ended experiment	6	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	2	2					
CO3	3	3					
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop Programs for Word Analysis		
M1.01	Install NLTK	2	Applying
M1.02	Develop a program to a. convert the given text to speech b. convert the audio file speech to text	2	Applying
M1.03	Implement Preprocessing of text (Tokenization, Filtration, Script Validation, Stop Word Removal, Stemming)	3	Applying
M1.04	Develop a program to calculate the term frequency of words in a given document.	2	Applying
M1.05	Develop programs to count and display the frequency of different part-of-speech categories (e.g., nouns, verbs, adjectives).	3	Applying
CO2	Develop programs for word generation		

M2.01	Write a Python program that generates a random word by combining a random selection of letters. (use the English alphabet for this task)	3	Applying
M2.02	Implement a simple Markov Chain-based word generator using a small dataset of words .	3	Applying
M2.03	Develop a python program to generate words based on a specific theme or constraint. (Use syllable patterns, phonemes, or other linguistic elements to generate unique and interesting words)	3	Applying
M2.05	Write a program to create, concatenate and print a string and access sub-string from a given string.	3	Applying
	Lab Exam – I	1.5	
CO3	Implement programs for Morphological Analysis		
M3.01	Write a Python program that takes a list of base words (lemmas) as input. Tokenize and apply both stemming and lemmatization to the words.	3	Applying
M3.02	Write a Python program that implements morphological analysis with a. stemming, b. synonym, c. stop word removal, d. regular expression	4	Applying
M3.03	Develop a program to identify Morphemes in tagged text and segmented text	3	Applying
M3.04	Develop a python program to implement parts of speech tagging	2	Applying
CO4	Open ended experiment		
M4.01	Open ended experiment**	15	Applying
	Lab Exam -2	1.5	

**** For example -Sentiment Analysis of Movie Reviews**

Text / Reference

T/R	Book Title/Author
T1	Tanveer Siddiqui, U.S. Tiwary, “Natural Language Processing and Information Retrieval”, Oxford University Press, 2008.
R1	Daniel Jurafsky and James H Martin, “Speech and Language Processing: An introduction to Natural Language Processing, Computational Linguistics and Speech Recognition”, 2 nd Edition, Prentice Hall, 2008.

Online Resources

Sl.No	Website Link
1	https://www.coursera.org/specializations/natural-language-processing